PriorityKV: Structure-Aware KV Retention with a Measurable Operating Boundary

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Abstract

A long agent conversation packs tool schemas, persistent identifiers, superseded instructions, and filler into one key–value (KV) cache. Every token costs the same amount of memory, but losing the wrong one costs far more than losing the wrong filler. A retention rule can read the structure a chat transcript already exposes—keep schemas, constraints, and identifiers before ordinary filler—and, unlike attention-derived selection, that signal costs nothing to compute. This paper asks when such a prior is sufficient, and answers it with a measured operating range rather than an assertion.

We sweep the protected-role mass of a controlled agent benchmark across eight levels (5% to 92%) crossed with three keep budgets, 15,360 scored generations on Qwen3-8B and Llama-3.1-8B. Structural retention holds at ceiling while the protected set fits the budget and degrades sharply once the oversubscription ratio $|S \cup M|/B$ passes one; the knee sits at ≈ 1.1 on both model families. Past it, structure loses to matched-budget SnapKV in 20 of 24 cells on Llama with zero wins in the opposite direction.

The collapse is not uniform across task types, and the exception identifies the real mechanism. State carried in the leading system block survives 18× oversubscription intact (tool-schema conformance stays at 1.000), because the evaluated policy breaks ties by position and that block is earliest. State carried in mid-conversation turns falls to exactly 0.000. The governing quantity is therefore not protected mass alone but its interaction with where load-bearing state sits relative to the policy's tiebreak—which explains directly why the same rule scores 0.000 on BFCL, whose schemas are in the system prompt while its task-critical state is not.

Because the ratio is computable before generation, we evaluate ADAPT, a budget-relative blend of structure and attention with no fitted parameter. ADAPT never loses to SnapKV in 48 cells across both models, and significantly exceeds it on Qwen at a 2% budget (up to $p = 4.9 \times 10^{-4}$, with no losing discordant pairs). That advantage does not replicate on Llama, where SnapKV never degrades enough to leave room. We report the scope honestly, together with a pre-registered external test whose pilot shows attention selection floors on BFCL below a 10% budget.

1 Introduction

Consider a long agent trace whose final tool call omits a required field. The field appeared in a schema many turns earlier, but its KV entries were evicted to meet a memory budget. Byte for byte, the schema looked no different from the filler that survived; operationally, losing it invalidated the call. The same failure can resurrect an obsolete instruction or substitute the wrong persistent identifier. Figure 1 illustrates this mismatch between storage cost and failure cost.

Autoregressive inference stores prior keys and values so each decode step can reuse the prefix. The cache grows with sequence length and constrains serving capacity [1, 2]. Existing work reduces that cost through paging, streaming windows, attention-derived eviction, or quantization [3, 4, 5, 6, 7]. Agent traces expose an additional signal: their serialization already identifies message roles, tool contracts, constraints, results, and state. PriorityKV asks whether this visible structure can guide what survives.

The answer is a useful contrast. Structure succeeds when important state is sparse and recognizable, then loses its ability to discriminate when nearly all tokens carry the same protected role. That reversal exposes the paper's main result: structural retention has an operating boundary that can be measured before generation. We then test whether a soft structure–attention blend can cross that boundary gracefully. A separate BF16/INT4 study tests storage rather than eviction; Figure 2 keeps these questions distinct.

The paper follows that contrast. Sections 3–4 define the controlled benchmark and policy; Section 5 establishes the in-distribution result and its placement, budget, and model limits; Section 7 presents the external crossover,

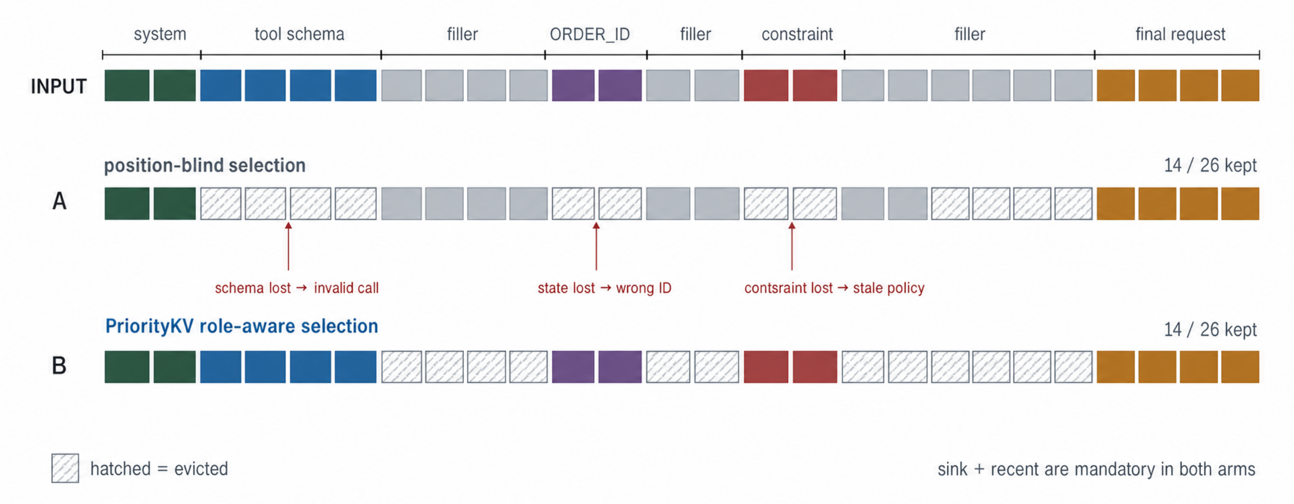

Figure 1: Equal storage cost, unequal failure cost. Two policies keep the same illustrative 14/26-token budget. Hatched cells are evicted; callouts show the resulting failure modes rather than empirical counts.

Study	Question and comparison	Measured endpoint	Interpretation
H1	**Retention · W5, n=120** Which positions survive at k=0.25?	112/120 vs 1/120	In-regime structural win
H2	**INT4 quality · W3, n=240** Does role-aware placement beat uniform?	0.8833 vs 0.8792	No quality separation
H3	**Systems · H200, n=18** Mixed FlashInfer vs FullKV/SDPA	0.719× / 1.200–1.215×	Payload–latency trade-off

All endpoints use fixed manifests and seeds.

Figure 2: Three questions, three verdicts. Eviction reliability, precision placement, and systems performance are evaluated independently; displayed counts and ratios are measured endpoints.

its mechanism, and ADAPT. The INT4 and systems results remain separate because eviction removes information whereas quantization keeps an approximation.

This work makes four contributions.

1. We introduce PriorityBench-A, a deterministic stress test for tool-schema conformance, instruction supersession, and state recall, with placement controls and a token-level retention audit.

2. We measure the operating range of structural retention as a dose–response curve rather than asserting it: eight protected-mass levels crossed with three keep budgets, on two model families, 15,360 scored generations. The boundary sits at oversubscription ratio ≈ 1 and replicates across architectures (Section 6).

3. We identify the mechanism by dissociating task types. Load-bearing state in the leading system block survives arbitrary oversubscription under a position-ordered tiebreak, while mid-conversation state does not. This supersedes a protected-mass-only account and explains the external BFCL result directly.

4. We evaluate ADAPT, a budget-relative structure–attention blend with no fitted parameter, and report both its significant advantage over SnapKV on Qwen at a 2% budget and its failure to replicate on Llama, together with a pre-registered external test and the pilot that scopes it.

2 Related Work

Paging and streaming. PagedAttention separates logical KV blocks from non-contiguous physical storage to reduce fragmentation and enable sharing; it does not choose semantic tokens to retain [2]. StreamingLLM identifies attention sinks and combines initial tokens with a recent window for streaming language modeling [3]. PriorityKV uses sink plus recent as a mandatory region and as the shape of its position-only control, not as a claim to reproduce StreamingLLM's streaming setting.

Eviction and attention-derived selection. H2O retains accumulated-attention heavy hitters together with recent tokens [4]. SnapKV uses an observation window near the end of the prompt to select clustered prefix features [5]. PyramidKV varies the cache budget across layers in response to pyramidal information funneling [6]. These methods infer importance from model behavior; PriorityKV tests application-visible message structure. The attention-baseline comparison uses repository reimplementations at matched token budgets, not the authors' reference code. In particular, H2O is a chunked SDPA reimplementation and carries an explicit fidelity caveat. A hybrid that protects structural tokens and fills the remainder with SnapKV was evaluated: it equals SnapKV on Qwen at 25% keep and is lower than both parents on the two Llama 5% slices. Thus the simplest union does not provide the proposed complementarity at the tested settings.

KV quantization and attention engines. KIVI motivates asymmetric group-wise low-bit KV quantization and keeps residual tokens at full precision [7]. PriorityKV's INT4 path is not a new quantizer and is not positioned as a replacement for KIVI. It is a concrete storage path for testing role-aware precision placement and accounting actual payload bytes. FlashInfer exposes composable attention kernels and log-sum-exp state merging for heterogeneous KV layouts [8]. PriorityKV uses that merge contract, while retaining an explicit BF16 cold scratch limitation.

Long-context evaluation. LongBench and RULER test broad long-context capabilities [9, 10]. Our suite instead targets exact failure attribution in synthetic agent protocols. It cannot establish workload prevalence or replace general long-context suites.

Agent and tool-use evaluation. BFCL adds stateful multi-turn and multi-step function-calling tasks [11], while τ-bench evaluates policy-constrained interaction with tools and a simulated user [12]. We use BFCL for external task success and public τ-bench trajectories for a generation-free retention audit. Neither benchmark was used to construct PriorityBench-A.

3 PriorityBench-A

3.1 Tasks and scoring

PriorityBench-A generates multi-turn chats in which an early span contains the state required by a final request and intervening turns provide benign filler. Scoring is binary and programmatic; no LLM judge is used. Table 1 summarizes the three equally represented categories in the locked 240-example manifest.

Table 1: PriorityBench-A task families in the locked manifest ($n = 240$; 80 examples per category).

Category	Required behavior	Deterministic scorer
Tool schema	Emit a valid call under an early contract	JSON/schema subset
Instruction supersession	Follow the latest conflicting constraint	Required/forbidden regex
Multi-turn state	Reuse an earlier ID, path, or preference	Exact slot match

Two frozen manifests serve different studies. A 240-example set covers nominal 8k, 16k, and 32k contexts for mixed-precision quality; a 120-example stress set covers 8k and 16k contexts for retention. Both balance the three task families. Frozen model revisions and manifest digests remain recorded in the public repository.

3.2 Placement and leakage controls

The generator's visible templates are also a threat: a heuristic tagger can appear semantic if gold state is consistently short, early, or marked by a template recognizable to the policy. We therefore evaluate two transformations of each stress slice. *Middle relocation* moves state away from the prefix while retaining the leading system message and final request. *Buried state* lengthens or embeds state-bearing turns so that a short-span heuristic does not protect all gold.

A separate CPU audit maps the gold-bearing messages into tokenizer spans and intersects them with the selected positions. This audit uses no model outputs. It measures the gold fraction already in sink plus recent and the gold fraction retained by each policy. Thus placement robustness and label leakage into the always-kept window are tested independently.

4 Method

4.1 Roles and matched-budget retention

Let a tokenized trace be $x_{1:n}$ with structural roles $r_{1:n}$. For keep fraction k, the nominal budget is

$$B(n,k) = \mathrm{round}(kn). \tag{1}$$

The mandatory set is $M = M_{\text{sink}} \cup M_{\text{recent}}$, where the first 16 tokens and last 128 tokens are protected. In the evaluated token-level regime, each compressed arm selects $A \supseteq M$ with

$$|A| = \max(B(n,k), |M|). \tag{2}$$

Position-blind retention uses the mandatory sink and fills from the recent edge. A corrected random control keeps a fixed M and samples the remaining budget; the released synthetic selector accidentally duplicates the position-blind indices, so those rows are collapsed in Section 5.1. Structure-aware retention first includes system, tool, constraint, and other state-bearing positions, then fills any residual budget from the recent edge. Whole-page experiments apply the same priority order while respecting page boundaries and never exceeding the token budget except for M.

The tagger consumes message roles and transparent schema or constraint markers. It does not inspect benchmark answers or future attention. The policy protects explicit protocol roles more reliably than unmarked free-form state; the buried controls in Section 5.1 quantify that boundary.

This ordering exposes a budget test. Let S denote structurally protected positions and retain the mandatory positional set M from above. If $|S \cup M| \le B$, every protected position fits and the structural score can separate it from filler. If $|S \cup M| > B$, the protected set is *oversubscribed*: a binary role label cannot order all protected positions, so the evaluated policy resolves the tie by position. We report the measured protected-role fraction as a cheap proxy for this condition; the fixed sink/recent contribution is explicit in M.

4.2 Pages and mixed precision

The physical page size is $P = 16$ tokens. In the mixed-precision study, every position remains present and an exact target fraction $f = 0.75$ is stored as INT4, subject to the common sink/recent BF16 protection. The position-blind arm spaces INT4 positions across the demotable region. The structure arm demotes filler, generated, and low-risk positions first, spilling into protected soft roles only if the byte budget requires it. Both arms therefore have the same number of INT4 positions.

Figure 3 traces the implemented path from tagging to the page table. It separates policy decisions—retention for eviction or dtype for mixed precision—from physical payload. Cold K and V pages use one uint8 container per 4-bit code plus per-group FP32 scale and zero-point metadata; hot pages remain BF16. Thus $0.719\times$ is the measured implementation payload, whereas the $0.473\times$ byte model assumes two codes per byte and lighter metadata.

4.3 Two-call FlashInfer decode

Prefill uses the native Hugging Face path. The resulting cache is partitioned into coalesced hot BF16 pages and uint8-coded cold pages. Before the frozen full-scratch decode measurement, cold pages are dequantized into BF16 GPU scratch. Each layer makes at most two homogeneous FlashInfer calls: one over hot pages plus the decode tail and one over cold scratch. Their outputs and native log-sum-exp states are combined with `merge_state`. We measured a maximum absolute error of approximately 4.88×10^{-4} after correcting an LSE-base mismatch.

Figure 4 shows why cache payload, allocator peak, and latency must be reported separately. The code-plus-metadata payload persists, full cold scratch is live during decode, and the decode tail grows token by token. A separate streamed-cold experiment instead dequantizes chunks and releases them after merging, but its three-example run is only a successful smoke test and is not a comparative systems result.

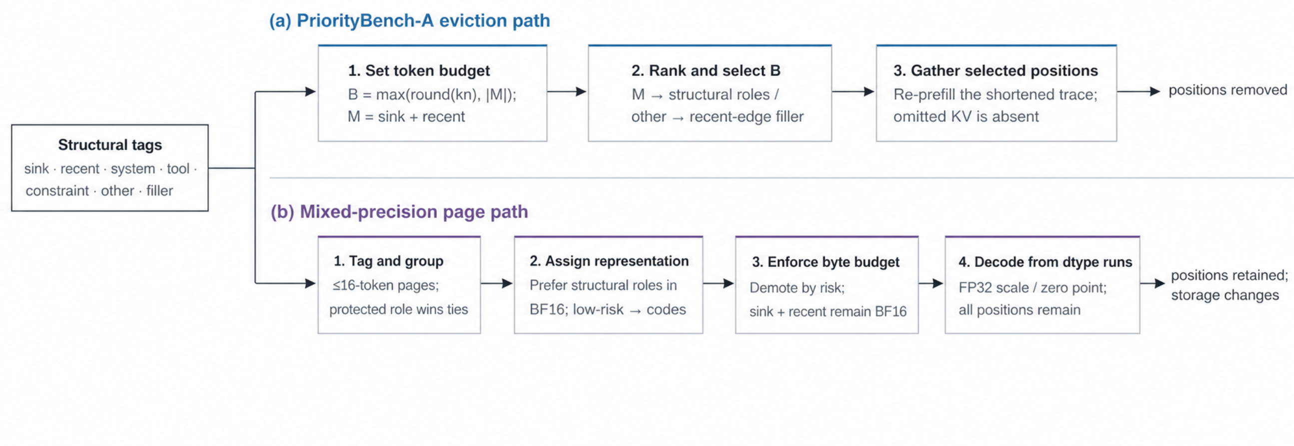

Figure 3: PriorityKV architecture. Chat metadata becomes token roles and pages of at most 16 tokens. Eviction and precision placement then use separate controllers and are evaluated independently.

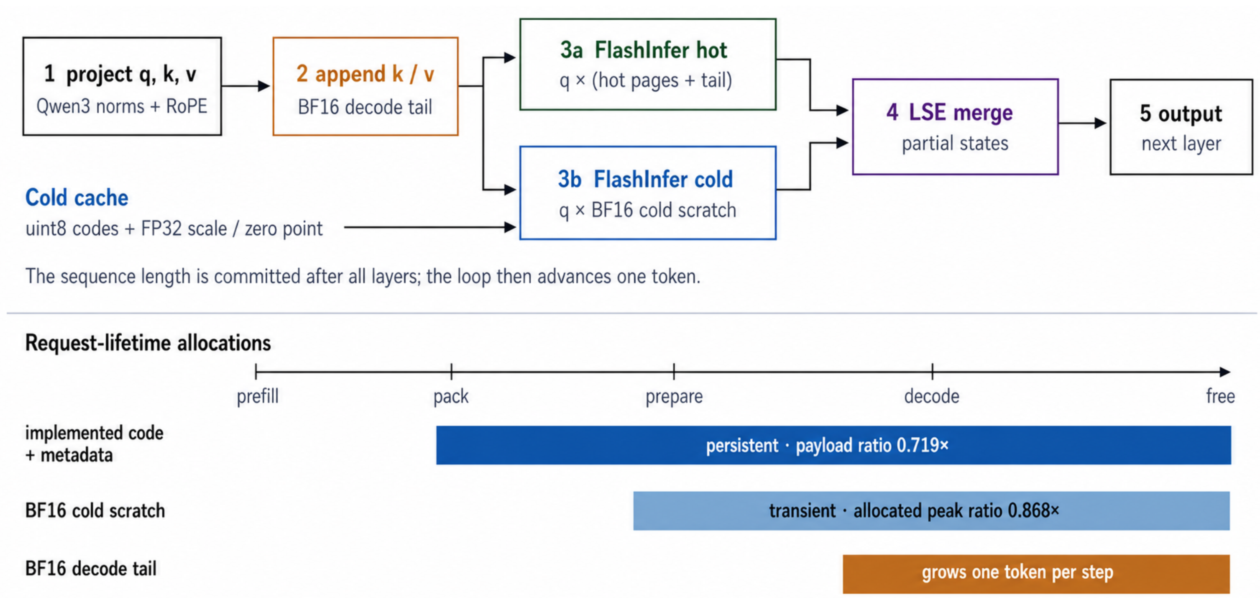

Figure 4: Decode path and allocation lifetime. Two homogeneous FlashInfer calls are merged through log-sum-exp state; the coded payload persists while BF16 cold scratch is live during decode.

5 Results

5.1 Structure protects state that blind eviction drops

At matched 25% token keep, structure passes 112/120 Qwen examples while the position-blind control passes 1/120 (Table 2). The corresponding Wilson intervals do not come close to overlapping. We report a single position-blind control rather than two: our released "random" arm is byte-identical to the uniform arm, a defect characterized in §9.1 rather than left for a reader to discover. Figure 5 places the position-blind and attention-derived selectors on one pass-rate axis.

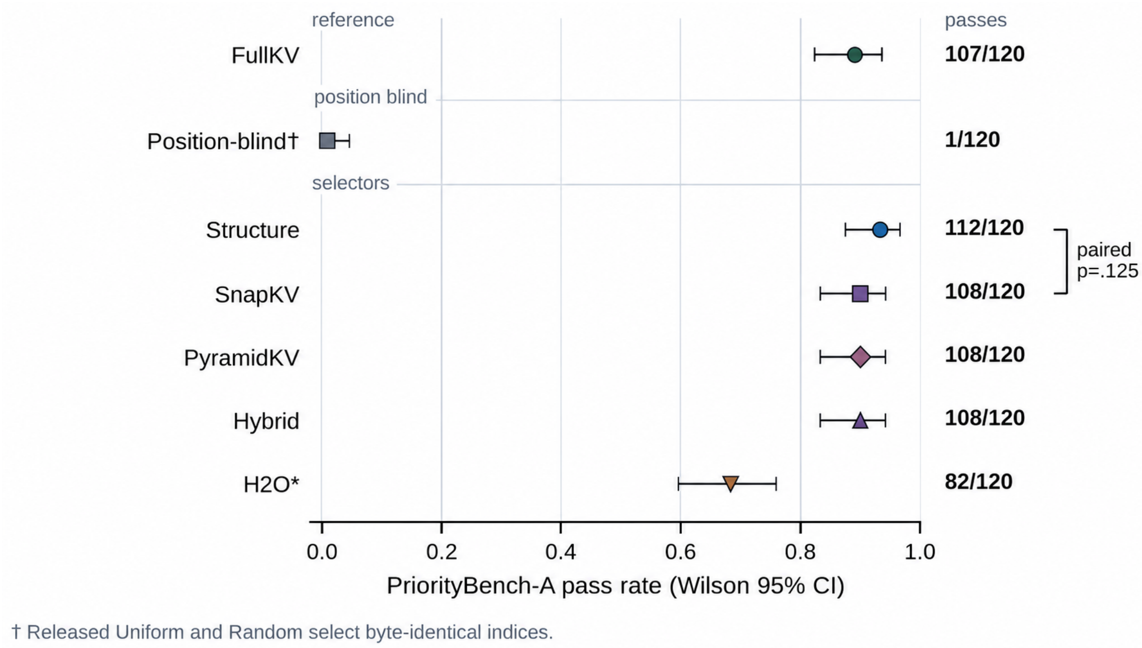

Figure 5: Matched-budget Qwen comparison ($n = 120$, $k = 0.25$). Points show pass rates with Wilson 95% intervals; exact counts are aligned at right and the bracket reports the paired structure–SnapKV McNemar test. H2O* denotes the chunked repository reimplementation.

Table 2: Pooled Qwen eviction and attention-baseline results at $k = 0.25$ ($n = 120$). Intervals are Wilson 95%.

Arm	Passes	Rate	Wilson 95% interval
FullKV	107/120	0.892	[0.823, 0.936]
Structure	112/120	0.933	[0.874, 0.966]
Position-blind	1/120	0.008	[0.002, 0.046]
SnapKV	108/120	0.900	[0.833, 0.942]
PyramidKV	108/120	0.900	[0.833, 0.942]
Hybrid	108/120	0.900	[0.833, 0.942]
H2O (chunked)	82/120	0.683	[0.596, 0.760]

The CPU audit confirms that this gap reflects retention rather than an easy always-kept window. Only 1.0–1.3% of Qwen gold tokens and none of the Llama gold tokens lie in sink plus recent (Table 3). Structure retains all mapped gold tokens, whereas uniform retains only the small fraction already in that window.

Placement controls bound the positive result (Table 4). Moving state to the middle gives both structure and FullKV 115/120, or 0.958 [0.906, 0.982]. Burying state in longer free-form turns lowers structure to 80/120, or 0.667 [0.578, 0.745], versus FullKV at 107/120, or 0.892 [0.823, 0.936]. Uniform scores 3/120 in the middle condition and 0/120 when state is buried. Visible roles are therefore a useful prior, not an oracle for arbitrarily embedded state.

Table 3: Gold-span retention audit at $k = 0.25$ ($n = 40$ per slice). Entries are mean token fractions, not pass counts.

Model	Slice	Gold in M	Structure	Uniform
Qwen	0	0.010	1.000	0.010
Qwen	1	0.013	1.000	0.013
Qwen	2	0.010	1.000	0.010
Llama	0	0.000	1.000	0.000
Llama	1	0.000	1.000	0.000
Llama	2	0.000	1.000	0.000

Table 4: Placement controls ($n = 40$ per slice), reported as structure / FullKV / uniform pass rates.

Slice	Middle relocation	Buried state
0	0.975 / 0.975 / 0.025	0.675 / 0.900 / 0.000
1	0.950 / 0.950 / 0.050	0.650 / 0.875 / 0.000
2	0.950 / 0.950 / 0.000	0.675 / 0.900 / 0.000

5.2 Structure reaches the attention-selector range in its sparse-state regime

PriorityBench-A establishes the sparse-state regime: the protected set fits inside the budget, and structure reaches the same measured range as attention selection. Section 7 probes the complementary schema-dense regime, where the structural score saturates. The protected fraction in Section 7.2 connects the two outcomes.

Structure passes 112/120, while SnapKV, PyramidKV, and hybrid each pass 108/120 (Table 2). The structure interval overlaps every 0.900 arm. The paired structure–SnapKV comparison has $b = 4$ cases where structure alone passes and $c = 0$ in the opposite direction; exact two-sided McNemar $p = 0.125$. The difference is not statistically significant. Hybrid equals SnapKV rather than exceeding it, so this simple union provides no additional benefit at the tested setting. H2O passes 82/120 using the chunked implementation; its implementation caveat prevents a reference-H2O claim.

5.3 Budget determines when the structural prior remains discriminative

All five tested structure and attention arms pass all 120 Llama examples at $k = 0.25$; the Wilson interval for 120/120 is [0.969, 1.000]. With no uniform arm, this is an easy-task ceiling rather than evidence of universal transfer. At $k = 0.05$, SnapKV passes 40/40 on each evaluated slice, while structure passes 35/40 and 36/40. No paired test was produced for these Llama slices.

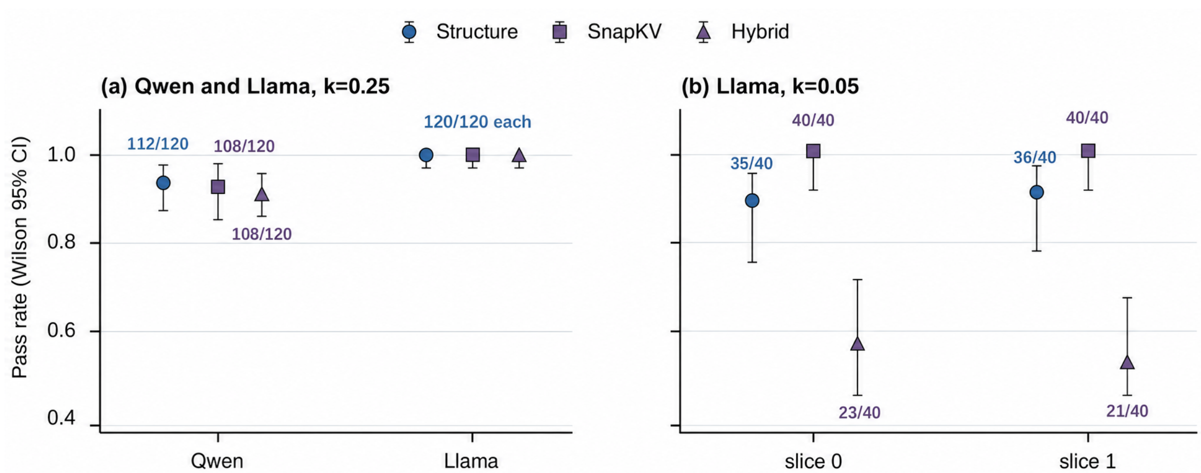

Figure 6: Budget and model dependence. Qwen and Llama at $k = 0.25$ use $n = 120$; each Llama $k = 0.05$ condition uses $n = 40$. Error bars are Wilson 95% intervals.

At the tighter budget, failures concentrate in tool-schema tasks. Structure passes 8/13 and 9/13 schema examples in the first and second slices, while SnapKV passes 13/13 in both. Hybrid is worse at 1/13 and 0/13, with overall rates 0.575 and 0.525. Protecting broad structural spans can therefore consume too much residual budget, and simply unioning role and attention selections does not make them complementary.

5.4 Role-aware and uniform INT4 placement preserve similar quality

On the locked 240 examples, FullKV passes 213, uniform-mixed 211, and structure-mixed 212 (Table 5). The structure–uniform difference is one example, and no paired statistical test supports a quality distinction. At 8k and 16k, every arm is 1.0; at 32k, FullKV, uniform, and structure are 49/76, 47/76, and 48/76. Figure 7 shows the count-backed Wilson intervals by context. The tested data do not support a role-aware INT4 quality advantage.

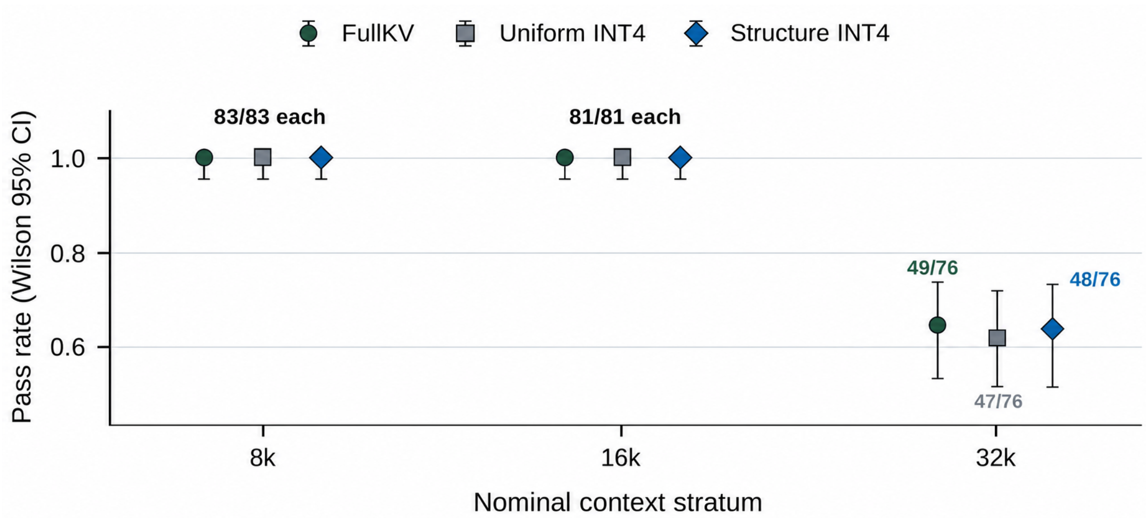

Figure 7: Matched INT4 quality. Qwen3-8B with FullKV or matched $f = 0.75$ INT4 placement. Error bars are Wilson 95% intervals from exact counts (83, 81, and 76 examples).

Table 5: Locked Qwen quality ($n = 240$). Intervals are Wilson 95%.

Arm	Passes	Rate	Wilson 95% interval
FullKV	213/240	0.8875	[0.841, 0.922]
Uniform mixed	211/240	0.8792	[0.832, 0.915]
Structure mixed	212/240	0.8833	[0.837, 0.918]

5.5 Storage savings incur scratch and latency costs

The mixed structure path has a measured code-plus-metadata payload ratio of 0.719× and an allocated-peak ratio of 0.868× FullKV on the 18-example memory slice. An idealized byte model gives 0.473× by assuming two INT4 codes per byte and lighter metadata; the current implementation instead stores each code in a uint8 container with FP32 scale and zero-point arrays. BF16 cold scratch further limits the peak-memory reduction. Figure 8 reports these memory and latency ratios. The canonical jobs were not independently repeated, so we do not show uncertainty bars.

At 8k, structure packing and scratch construction take 34.8 and 14.2 ms. E2E TTFT is 1.118× FullKV, and TPOT rises from 25.25 to 30.29 ms (1.200×). At 16k, pack and scratch take 48.1 and 19.7 ms; E2E is 1.113×, and TPOT rises from 23.84 to 28.95 ms (1.215×). This is the measured latency cost associated with the payload savings. The streamed-cold smoke run exits successfully with parity and approximately 36.4 GiB peak in its log, but with only three examples and no comparative frontier it remains a smoke test rather than a systems result.

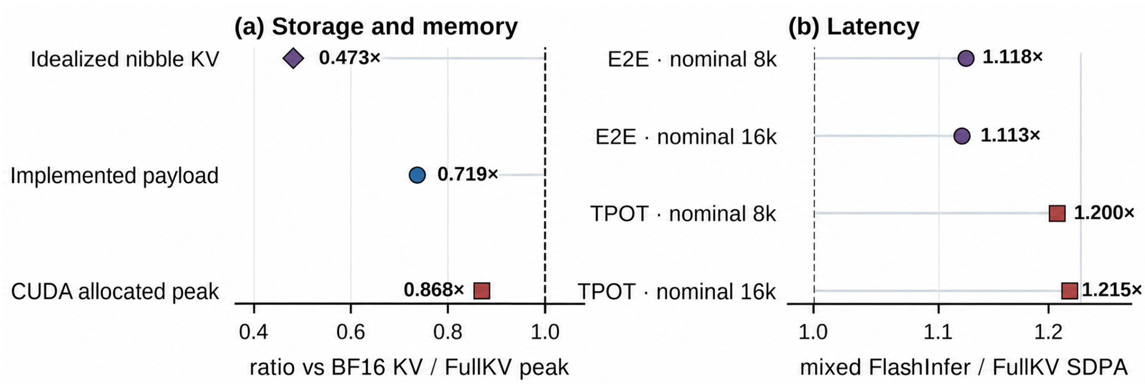

Figure 8: H200 systems trade-off. The implemented code-plus-metadata payload is smaller than FullKV KV, while the allocated CUDA peak falls less. Latency compares mixed FlashInfer against FullKV SDPA at nominal context strata; the dashed line marks parity.

6 The Operating Boundary, Measured

Sections 5.1–5.3 report two points on an axis: PriorityBench-A with sparse protected mass, and (in Section 7) BFCL with nearly saturated protected mass. Two points do not establish a boundary. This section replaces them with a controlled sweep.

6.1 Design

We parameterise the generator so that filler user/assistant pairs can be converted into *distractor* tool-schema pairs: a short registry notice and a JSON dump of deferred schemas, sized to the characters they replace. The gold task, its scoring payload, and the approximate context length are unchanged; only the composition of the surrounding context varies. Eight levels span 5.2%–92.2% measured protected-role mass, crossed with keep budgets $k \in \{0.02, 0.05, 0.10\}$ at 16k nominal context. The same base example appears at every level, so comparisons along the axis are paired. Every non-FullKV arm is a `kvpress` press over an identical full prefill, as in Section 7.

Two design iterations are reported because both were informative. A first sweep placed gold state in the template's natural prefix position; there every non-blind arm scored 1.000 in all 24 cells, including at 9× oversubscription. The cause is the tiebreak: when the protected set does not fit, the evaluated policy retains the *earliest* protected positions, which in that layout are the gold positions themselves. A second iteration relocated gold to mid-context but applied the transform after schema conversion; because the relocation helper classifies converted schema turns as non-filler, gold drifted back toward the prefix as protected mass rose (character fraction 0.65 at the lowest level, 0.07 at the highest), cancelling the effect. The reported sweep applies relocation *before* conversion, with gold verified at character fraction 0.63–0.65 at every level.

6.2 The boundary and its replication

Table 6 reports structure against the oversubscription ratio $|S \cup M|/B$. Structure is at ceiling while the protected set fits and falls immediately past ≈ 1.1 on both models. FullKV is 1.000 in every cell, so the collapse reflects retention rather than task difficulty; position-blind arms remain near zero throughout.

Table 6: Structure pass rate against oversubscription ratio, pooled across budgets ($n = 60$ per cell). SnapKV and ADAPT are shown at the same cells.

Ratio	Structure (Qwen)	Structure (Llama)	SnapKV	ADAPT
0.5×	1.000	1.000	1.000	1.000
1.0×	1.000	1.000	1.000	1.000
1.1×	1.000	0.883	1.000	1.000
2.2×	0.617	0.600	1.000	1.000
4×–18×	0.400	0.333–0.417	0.967–1.000	1.000
27×–45×	0.067	0.000	0.750–0.867	0.950–1.000

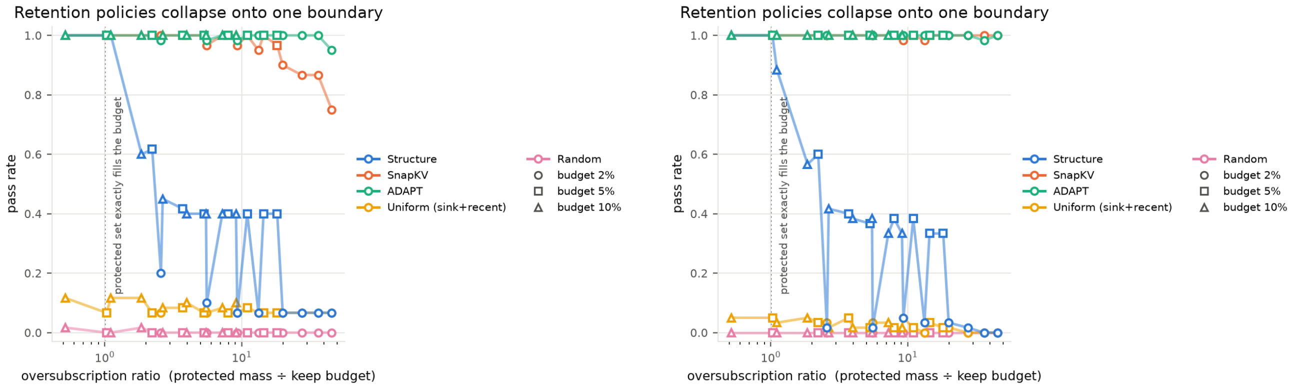

Figure 9: One boundary, two architectures. Pass rate against oversubscription ratio for Qwen3-8B (left) and Llama-3.1-8B (right). Marker shape encodes the keep budget, so points from 2%, 5% and 10% budgets are visibly on the same curve. The dotted line marks the protected set exactly filling the budget.

Past the boundary the ordering against attention selection reverses relative to Section 5.2. On Llama, structure loses to matched-budget SnapKV in 20 of 24 cells with $b = 0$ in every one; on Qwen the same pattern holds at the tightest budget ($b = 0$, $c = 41$–54). The parity reported at $k = 0.25$ on PriorityBench-A is therefore a property of that operating point, not a general result.

6.3 Which state survives, and why

Pooling across task families conceals the mechanism. Table 7 splits it. At high oversubscription, tool-schema conformance is perfectly retained while instruction supersession is at exactly zero.

Table 7: Structure by task family at high oversubscription ($\geq 5\times$; $n = 20$ per cell per model).

Model	Budget	Tool schema	Supersession	Multi-turn state
Qwen	0.10	1.000	0.000	0.200
Qwen	0.05	1.000	0.000	0.200
Qwen	0.02	0.000	0.000	0.200
Llama	0.10	1.000	0.000	0.000–0.150
Llama	0.02	0.000	0.000	0.000–0.050

The explanation is structural. A tool-schema item's load-bearing content is the leading system block, and a position-ordered tiebreak retains the earliest protected positions; the contract therefore survives however oversubscribed the policy is, until the budget is too small to hold the block at all (the $k = 0.02$ row). Supersession and multi-turn state live in mid-conversation turns, which the same tiebreak discards first.

This supersedes the account in Section 7.2. Protected mass relative to budget determines *whether* the label can still discriminate; the position of load-bearing state relative to the tiebreak determines *what is lost when it cannot*. The two together predict the external result of Section 7: BFCL prompts carry their schemas in the system block, which structure retains, while the state the task actually depends on accumulates across turns, which it does not.

6.4 ADAPT across the range

ADAPT (Equation 3) holds 0.950–1.000 on Qwen and 0.983–1.000 on Llama across all 24 cells per model, and loses no discordant pair to SnapKV anywhere in the 48 cells. Its measured α decays monotonically from 1.000 to 0.022 as oversubscription grows, with no value fitted to any outcome.

On Qwen at $k = 0.02$ it significantly exceeds SnapKV in four cells: 1.000 versus 0.900 ($p = 0.031$), 1.000 versus 0.867 ($p = 7.8 \times 10^{-3}$, twice), and 0.950 versus 0.750 ($p = 4.9 \times 10^{-4}$), with $c = 0$ throughout. The advantage does not replicate on Llama, where SnapKV does not fall below 0.983 at any tested setting and therefore leaves no headroom. We claim only that ADAPT matches attention selection everywhere tested and exceeds it where attention selection begins to degrade.

7 External Evaluation: Finding the Boundary

PriorityBench-A shows that visible structure can protect sparse state. We next ask when that prior stops being informative. BFCL V3 multi-turn provides a sharply different, externally authored context distribution [11]. We score it with the unmodified official `multi_turn_checker`, which executes tool calls against stateful API implementations and compares final state and execution responses against ground truth. All non-FullKV arms are `kvpress` presses over an identical full prefill at an identical compression ratio, so the arms differ only in which KV entries survive, never in how they are removed.

7.1 BFCL exposes the high-protected-mass regime

We start with 150 conversations per model at a 25% keep budget. After paired context-limit exclusions, 140 Qwen and 143 Llama conversations remain (Table 8).

Table 8: BFCL V3 multi-turn, official scorer, 25% keep. Paired completeness 0.933 (Qwen) and 0.953 (Llama); exclusions are all `MODEL_CONTEXT_LIMIT`; zero matched-budget violations.

Arm	Qwen (n=140)	Llama (n=143)
FullKV	0.193	0.077
SnapKV	0.136	0.084
ADAPT (§7.3)	0.129	—
Structure	0.000	0.000
Uniform	0.000	0.000
Random	0.000	0.000

The low FullKV scores, 0.193 and 0.077, make harness validation essential. Replayed ground truth passes the upstream checker, while corrupted arguments and dropped turns fail, and the zero-arm Wilson upper bound (0.027) sits below FullKV on both models. The reported runs use a corrected pipeline; §9.1 documents this fix alongside the other baseline correction.

This schema-dense regime locates the opposite end of the operating range: the binary structural score has no residual ranking signal and records 0/140 on Qwen and 0/143 on Llama. Exact paired McNemar gives FullKV versus structure $p = 1.5 \times 10^{-8}$ (Qwen) and $p = 9.8 \times 10^{-4}$ (Llama). Attention selection, by contrast, is not significantly different from FullKV at a 4× smaller cache: $p = 0.152$ on Qwen and $p = 1.0$ on Llama. Structure versus SnapKV is significant on both ($p = 3.8 \times 10^{-6}$ and $p = 4.9 \times 10^{-4}$). The central pattern—a hard structural prior exhausts its signal while attention remains useful—thus replicates across two model families.

7.2 Protected-set oversubscription explains the crossover

The crossover has a measurable mechanism. A binary role score separates protected tokens from filler only while the protected set fits inside the keep budget. Once oversubscribed, the evaluated policy must choose among equally protected tokens using its positional tie-break. Figure 10 measures the protected-role fraction with the same tagger used by the policy.

The two datasets therefore test different sides of the same rule. PriorityBench-A contains distractor filler by construction, the regime in which "protect the structure, drop the filler" is effective. The sampled BFCL prompts contain little analogous filler, so broad role protection cannot discriminate.

A generation-free audit of 4,856 public τ-bench trajectories (828,000 extracted spans) corroborates the mechanism from the other side: structure retains explicit policy lines at 0.820 versus 0.001 for a position-only budget, but loses on reused identifiers (0.055 versus 0.315). Structure wins precisely where protected spans are scarce and separable.

7.3 ADAPT softens the prior beyond the boundary

The hybrid evaluated in §5.3 force-protects structural positions, which consumes the entire budget whenever protected mass exceeds it. That is a mis-specification rather than a dead end. ADAPT replaces the hard constraint

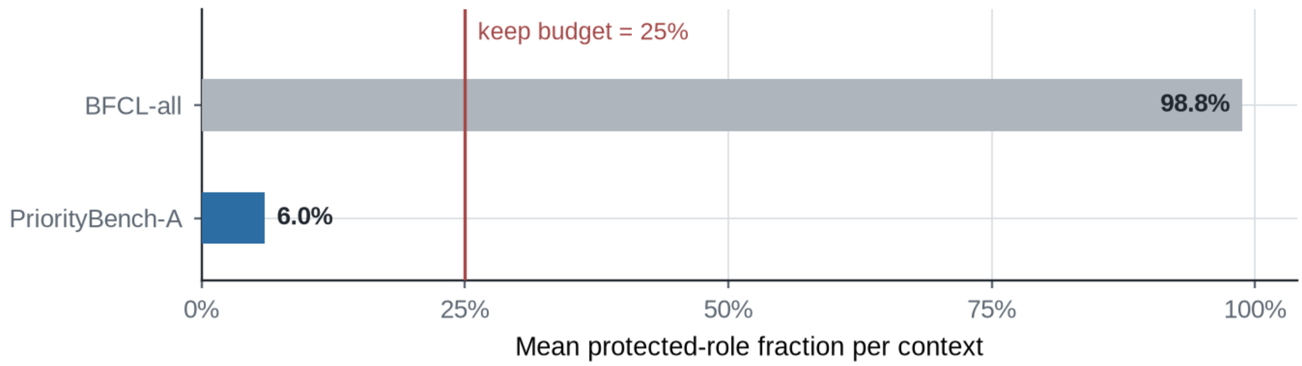

Figure 10: Protected mass relative to the keep budget. PriorityBench-A stress set has 6.0% mean protected-role mass ($n = 120$; 0/120 contexts exceed the budget), leaving room within a 25% budget. The sampled BFCL contexts are dominated by 32 JSON tool schemas and have 98.8% protected mass, so the role signal is oversubscribed. These two task-success workloads establish the observed crossover; broader validation remains future work.

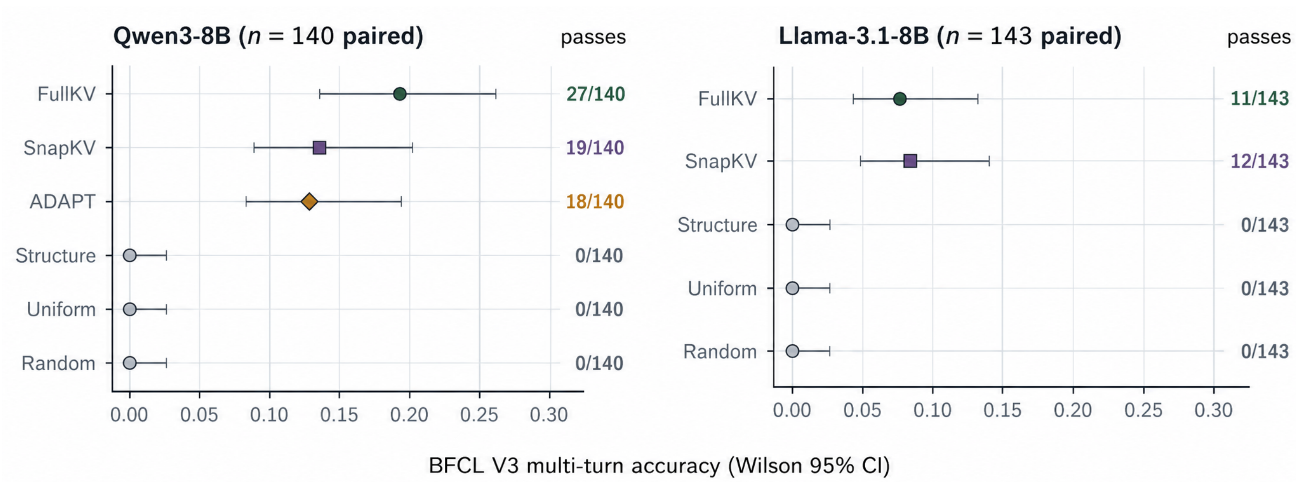

Figure 11: Paired BFCL accuracy with Wilson intervals. The same ordering holds on both models: no significant FullKV–SnapKV difference is detected, while the three non-attention policies are at zero with a Wilson upper bound of 0.027.

with a budget-relative prior,

$$\alpha = \min\left(1, \frac{\text{keep budget}}{\text{protected mass}}\right), \qquad s = \alpha \cdot \text{rank}(s_{\text{struct}}) + (1 - \alpha) \cdot \text{rank}(s_{\text{attn}}), \tag{3}$$

where both quantities are known from the prompt and no scalar is fitted to task outcomes. Rank normalization makes the unlike score scales comparable. When $\alpha = 1$, ADAPT recovers the evaluated structure ordering exactly; as oversubscription grows, it assigns increasing weight to attention. It does not become pure SnapKV at the measured 25% boundary.

On BFCL the measured α is 0.267 on average (range 0.250–0.401) over 833 generation steps. ADAPT scores 0.129 versus 0.136 for SnapKV (exact McNemar $p = 1.0$, $\Delta = -0.007$, 95% CI $[-0.057, +0.071]$) and has no significant paired difference from FullKV ($p = 0.108$); it significantly exceeds the three non-attention policies ($p = 7.6 \times 10^{-6}$). This result shows recovery to the attention-based range on this workload, not superiority to SnapKV. The formula was chosen before the recorded ADAPT run but committed afterwards, so we treat the experiment as exploratory rather than preregistered evidence.

8 Discussion

The boundary turns structure into a deployable prior, with a caveat the sweep makes explicit. Visible protocol structure protects sparse, recognizable state that positional selection drops, and the relevant test is whether $|S \cup M|$ exceeds the keep budget—a quantity available before generation. But Section 6.3 shows the ratio is not sufficient on its own. Because the evaluated policy resolves ties by position, state in the leading system block is retained at any ratio, while mid-conversation state is dropped as soon as the ratio exceeds one. A serving policy that consults the ratio alone will therefore be over-optimistic for workloads whose critical state accumulates across turns, which is precisely the agent setting this work targets. The deployable diagnostic is the pair: the ratio, and whether the state the task depends on lives where the tiebreak protects it.

Soft weighting is safer than a hard union. The original structure–SnapKV union equals SnapKV on Qwen and falls below both parents on Llama at 5%, because protected positions can consume the residual budget. ADAPT addresses that mechanism directly: it preserves the structural ordering when the set fits and shifts weight toward attention as oversubscription grows. Its BFCL result supports this recovery mechanism, while remaining exploratory and limited to one budget.

Bytes, peak, and speed are different systems quantities. Uint8-coded values and scale metadata reduce payload. Full-layer BF16 cold scratch then expands the live working set and adds preparation and decode cost. The observed 0.719× payload, 0.868× peak, and roughly 1.20× TPOT are mutually consistent. They do not establish serving throughput or capacity under concurrency.

9 Limitations and Threats to Validity

Synthetic workload. PriorityBench-A is generated from controlled templates and exact scorers. It measures whether particular protocol state survives, not how often these failures occur in deployed traffic. Its 6.0% mean protected-role mass places it at one end of the measured boundary; BFCL tests the other end.

External evaluation scope. The external results cover one budget, two 8B models, and 150 starting conversations per model. FullKV scores 0.193 and 0.077, leaving limited absolute headroom even though the arms separate. The τ-bench audit measures retention only—no simulator, backend, or generation—and excludes SnapKV, whose scores require realized attention.

Tagger–generator alignment. The tagger uses role, length, and schema/constraint markers that resemble the generator's visible templates. The gold audit rules out answer labels already in sink plus recent, but it does not eliminate all template alignment. The buried result—0.675, 0.650, and 0.675 versus FullKV 0.900, 0.875, and 0.900—is the clearest measured boundary.

9.1 Corrections

Two implementation bugs were caught and fixed before the reported runs; we document both here rather than leave a reader to find them. First, the released synthetic study listed "uniform" and "random" as separate controls, but `select_random` expands its forced recent window until the budget is full, leaving nothing to sample. We verified byte-identical index sets at $n \in \{600, 4096, 40000\}$ and therefore collapse the two 1/120 columns into one position-blind result; no measured score changes. The BFCL study uses a corrected random arm with a fixed recent window and independently scores 0.000. Second, an early BFCL run mis-decoded reasoning blocks, which forced every arm, including FullKV, to zero; the BFCL results in §7 use the pipeline after that fix.

Baseline fidelity. SnapKV, PyramidKV, H2O, and hybrid are matched-budget repository reimplementations, not reference-code leaderboard runs; chunked H2O is the least faithful approximation.

Statistical scope. The paired Qwen structure–SnapKV result at $k = 0.25$ is not significant ($p = 0.125$), and the Llama 5% comparison in Section 5.3 has no paired test. Canonical runs use fixed example sets rather than machine replications; Wilson intervals describe pass counts, not hardware variability.

Sweep scope and construction. The sweep is synthetic, and its protected-mass axis is built from distractor schemas authored to carry the same roles the tagger recognises, so the instrument and the stimulus are not independent; the axis is measured post hoc with the pinned tokenizer, and the mechanism replicates across two model families, but this does not remove the concern. FullKV scores 1.000 in every sweep cell, so the benchmark isolates retention cleanly but has no difficulty gradient and cannot speak to workload realism. The ADAPT advantage over SnapKV is confined to Qwen at $k = 0.02$; on Llama SnapKV never degrades enough to leave headroom, so that comparison is uninformative rather than negative.

External test status. A pre-registered tight-budget test of ADAPT on BFCL is specified in the repository's pre-registration note, committed before any tight-budget conversation was scored. Its pilot finds that matched-budget SnapKV floors on BFCL below a 10% keep budget (1/48, then 0/49 at 5% and 2%), against a FullKV reference of 9/48 on the same conversations. The regime in which the sweep predicts an ADAPT advantage therefore may not exist on this workload; the outcome of that test is not folded into any claim here.

Model and budget scope. Qwen3-8B on H200 is the primary matrix. Llama provides a ceiling and a tighter-budget crossover, not universal transfer. Only selected 5% and 25% token budgets are tested.

Systems scope. The latency comparison uses different backends (FullKV SDPA and mixed FlashInfer) and single-request measurements. It does not establish batched throughput, concurrency, or tail latency. Payload is not peak memory because cold attention expands the coded cache into BF16 scratch.

10 Conclusion

Application-visible structure is a retention signal that costs nothing to compute, and it has an operating range that can be measured rather than assumed. Across 15,360 scored generations on two model families, structural retention holds at ceiling while the protected set fits the keep budget and degrades sharply past an oversubscription ratio of one; past that point it loses to matched-budget attention selection with no wins in the opposite direction. The sweep also shows why a ratio-only account is incomplete: under a position-ordered tiebreak, state in the leading system block survives 18× oversubscription while mid-conversation state is lost entirely, which is the reason the same rule fails on externally authored agent traces whose schemas sit in the system prompt. ADAPT, whose single weight is read off the prompt with nothing fitted, matches attention selection in all 48 cells tested and exceeds it where attention selection itself begins to degrade. In operational terms: retain the schema before the pleasantry, but check first whether the budget can hold the schema—and whether the state your task depends on is the schema at all.

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